

THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF PHYSICAL SCIENCES
DEPARTMENT OF MATHEMATICAL SCIENCES
MTS315 - ENGINEERING MATHEMATICS I - CLASS TEST 2023/2024

Instructions: Answer all questions

Time: 30mins.

1. Find the Wronskian $W[y_1, y_2, y_3](x)$, where $y_1(x) = x$, $y_2(x) = x^2$ and $y_3(x) = x^{-1}$ are solutions to $x^3 y''' + x^2 y'' - 2xy' + 2y = 0$, $x > 0$. Hence find the general solution.
2. Determine the inverse Laplace Transform of $f(p) = \frac{5p^2 - 4p - 7}{(p-3)(p^2 + 4)}$
3. State and prove the superposition principle for the solutions of non-homogenous ordinary differential equations.

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B. TECH DEGREE EXAMINATION
FIRST SEMESTER 2023/2024
MTS315 - ENGINEERING MATHEMATICS I

Instruction: Answer any **four** questions.

Time allowed: 2 hrs

- 1a) ✓ Show that the function $y = Axe^x$ is the solution to the differential equation $x \frac{dy}{dx} - y(1+x) = 0$
- b) Find the Wronskian $W[y_1, y_2, y_3](x)$, where $y_1(x) = x$, $y_2(x) = x^2$ and $y_3(x) = x^{-1}$ are solutions to $x^3 y''' + x^2 y'' - 2xy' + 2y = 0$, $x > 0$. Hence find the general solution.
- 2a) ✓ Solve (i) $\frac{dy}{dx} = \frac{y-1}{x+3}$; $y(-1) = 0$ (ii) $\frac{dy}{dx} = \frac{\tan y}{x^2 - 1}$
- b) Show that $(4t^3 y^2 - 6t^2 y - 2t - 3)dt + (2t^4 y - 2t^3)dy = 0$, $y(0) = 3$ is exact and solve the equation
- 3a) ✓ Solve the differential equation $(1+x^2) \frac{dy}{dx} - \arctan x + y = 0$
- b) Determine the inverse Laplace Transform of $f(p) = \frac{5p^2 - 4p - 7}{(p-3)(p^2+4)}$
- 4a) ✓ Given that $L\{y(t)\} = \int_0^\infty y(t)e^{-st} dt = y(s) = \bar{y}$. Prove that $L\{y''(t)\} = s^2 \bar{y} - sy(0) - y'(0)$.
Hence, solve the equation $y'' + 2y' + 5y = e^{-t} \sin t$, $y(0) = 0$ and $y'(0) = 1$
- b) Solve $2xdy = (x+y)dx$
- 5a) ✓ Show that the following sets of functions are linearly independent
(i) x, x^2, x^3 (ii) e^x, e^{2x}, e^{4x}
- b) Show that the equation $y'' + y' - 6y = 0$ has two distinct solutions of the form $y = e^{ax}$.
- 6a) Superposition principle states that if y_1 is a solution to the equation $ay_1'' + by_1' + cy_1 = f_1(t)$ and y_2 is a solution to $ay_2'' + by_2' + cy_2 = f_2(t)$, then for any constants k_1 and k_2 , the function $k_1 y_1 + k_2 y_2$ is a solution to the differential equation $ay'' + by' + cy = k_1 f_1(t) + k_2 f_2(t)$. Prove and explain the theorem.
- b) Use the method of variation of parameters to obtain the solution of $4y'' - 4y' - 8y = 8e^{-t}$